Subspaces of $\mathbb{R}^n$ (hyperplane)

↳ lines, origin, plane, etc.

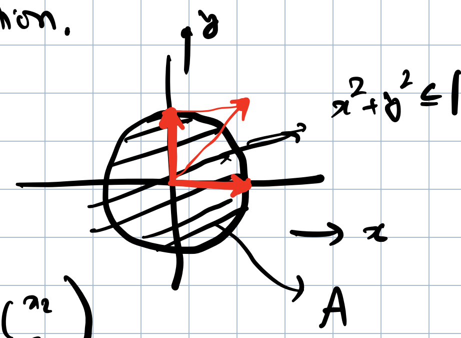
A set $A \subseteq \mathbb{R}^n$ is called a subspace if

for every $x, y \in A$, $\underbrace{\alpha \underline{x} + \beta \underline{y}}_{} \in A$ for any $\alpha, \beta \in \mathbb{R}$.

↳ linear combination.

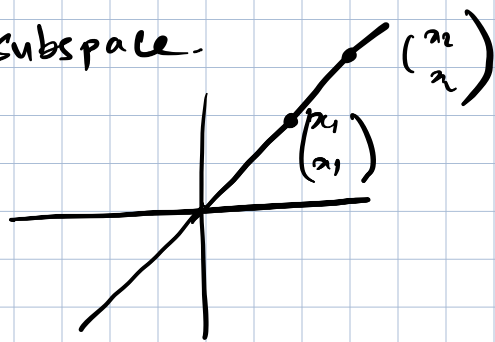
for any $\underline{x} \in A$, $\alpha \underline{x} \in A$.

Is A a subspace?



Examples: $A = \{\underline{0}\}$ is a subspace.

$$A = \left\{ \begin{bmatrix} x \\ x \end{bmatrix} : x \in \mathbb{R} \right\}$$



$$\begin{pmatrix} \alpha x_1 + \beta x_2 \\ \alpha x_1 + \beta x_2 \end{pmatrix}$$

the image & null space of a matrix are subspaces.

$$n(A) = \{ \underline{x} \mid A\underline{x} = 0 \}$$

$A\underline{x} = 0$

$A\underline{y} = 0$

Let $\underline{x}, \underline{y} \in n(A)$. Then

$$A(\alpha \underline{x} + \beta \underline{y}) = \alpha A\underline{x} + \beta A\underline{y}$$

$$= \alpha \underline{0} + \beta \underline{0} = \underline{0}$$

Hence $\alpha \underline{x} + \beta \underline{y} \in n(A)$. So $n(A)$ is a subspace.

$$im(A) = \{ A\underline{x} \mid \underline{x} \in \mathbb{R}^n \}$$

$\underline{y}_1 = A\underline{x}_1$

Let $\underline{y}_1, \underline{y}_2 \in im(A)$.

$\underline{y}_2 = A\underline{x}_2$

$$\alpha \underline{y}_1 + \beta \underline{y}_2 = A(\alpha \underline{x}_1 + \beta \underline{x}_2)$$

Hence $\alpha \underline{y}_1 + \beta \underline{y}_2 \in \mathrm{im}(A)$. Therefore im(A) is abi a subspace.

Consider the x, y plane.

this is the null space of the matrix

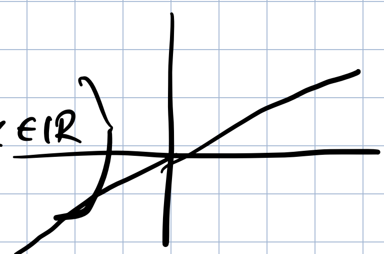
$$\underline{\underline{A}} = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \qquad n(A) = \left\{ \underbrace{\begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{z=0} = 0 \right.$$

Consider $B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$

$$\mathrm{im}(B) = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \right\} = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \right\}$$

<u>Experiment</u> Given a subspace, what is the largest number of LI vectors it can hold?

$$A = \left\{ \begin{pmatrix} x \\ x \end{pmatrix} : x \in \mathbb{R} \right\}$$



<u>Attempted solution</u>. $S = \emptyset$

"Greedy algorithms"

Pick any non-zero $\underline{x}_1 \in A$.

$$S \leftarrow S \cup \{\underline{x}_1\}$$

$\boxed{\text{Set produced may change}}$ everytime this algo is run

$\boxed{\text{Size of}}$ final set is same

Pick any vector $\underline{x}_2 \in A$ that is linearly indep of $\underline{x}_1$.

every time.

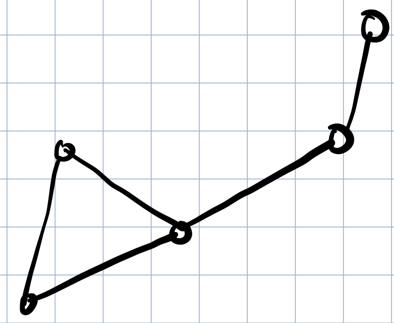
$$S \leftarrow S \cup \{x_2\}$$

Continue — —

pick $x_k \in A$ such that x_k is linearly indep of $x_1, x_2, \ldots, x_{k-1}$.

$$S \leftarrow S \cup \{x_k\}$$

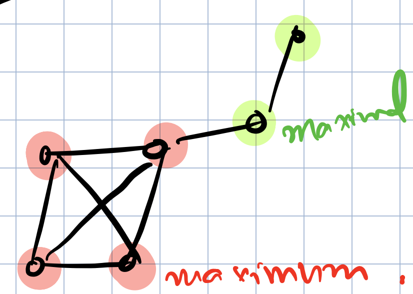
Stop/exit whenever such a cannot be found.



<u>Detour</u>: cliques in graphs

Find the largest clique in the graph!

Greedy approaches in general may not work.



maximal $\rightarrow$ set which cannot grow any further without destroying the given property.

maximum. $\rightarrow$ largest possible set of given property

Attempted solution above constructs a maximal LI set (which need not be maximum).

<u>Theorem</u>: Any maximal LI set is maximum.

<u>Proof</u>: (By contradiction) $k = 2$

$$\underbrace{x_1, x_2, \ldots, x_k}$$

$(k' > k)$

maximal

$\underline{y}_1 \quad \underline{y}_2 \quad \cdots \quad \underline{y}_k, \underline{y}_{k+1}, \cdots, \underline{y}_{k'} \leftarrow$ maximum.

$k' = 3$

$\underline{y}_1 = \alpha_{11}\underline{x}_1 + \alpha_{12}\underline{x}_2 + \cdots + \alpha_{1k}\underline{x}_k$

$\underline{y}_2 = \alpha_{21}\underline{x}_1 + \alpha_{22}\underline{x}_2 + \cdots + \alpha_{2k}\underline{x}_k$

$\vdots$

$y_{k'} = \alpha_{k'1}\underline{x}_1 + \alpha_{k'2}\underline{x}_2 + \cdots + \alpha_{k'k}x_k$

$$Y = [\underline{x}_1 \quad \underline{x}_2 \quad \cdots \quad \underline{x}_k] \underbrace{\begin{bmatrix} \alpha_{11} & \alpha_{21} & \alpha_{k'1} \\ \alpha_{12} & \alpha_{22} & \\ \vdots & \vdots & \\ \alpha_{1k} & \alpha_{2k} & \alpha_{k'k} \end{bmatrix}}_{k \times k'}$$

$\underline{\gamma}$

M

Fat / wide matrix

So we have $Y = XM$

for a wide matrix M. we know that

$\exists\, \underline{\gamma} \neq 0$ such that $M\underline{\gamma} = 0$.

Now, $Y\underline{\gamma} = XM\underline{\gamma} = 0$

therefore the columns of Y are linearly dependent,

a contradiction.

Corollary: the Attempted soln above always gets stuck

at the same size.

Dimension: Dimension of a subspace is the size of the largest linearly independent set in the subspace

largest $\longrightarrow$ maximum / maximal.

The dimension of a subspace is the size of the largest LI set in the subspace.

Basis of a subspace is a maximum / maximal LI set in the subspace.

dimension $\Longleftrightarrow$ degrees of freedom.

Spanning sets

$$\text{im}(A) = \{ A\underline{x} : x \in \mathbb{R}^n \}$$

$$\downarrow$$

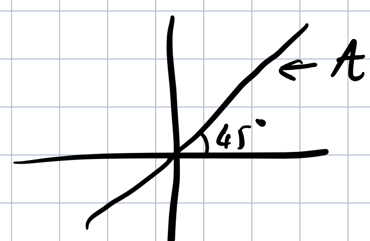
$$\underline{a_1} x_1 + \underline{a_2} x_2 + \cdots + \underline{a_n} x_n$$

Observations on im(A).

Let $\underline{a_1}$ be the first column of A.

Is $\underline{a_1} \in \text{im}(A)$? Yes

ith column $\longrightarrow \underline{a_i} \in \text{im}(A)$



$$x \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

$$\nearrow$$

$$x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

We say that S is a spanning set for a subspace A if every $\underline{v} \in A$ can be written as

$$\underline{v} = x_1 \underline{v}_1 + x_2 \underline{v}_2 + x_3 \underline{v}_3 + \cdots$$

where $S = \{\underline{v}_1, \underline{v}_2, \underline{v}_3, \cdots\}$

Example

$A = \mathbb{R}^2$

$S = \{\underline{e}_1, \underline{e}_2\}$

$\mathbb{R}^n$:

$\{\underline{e}_1, \underline{e}_2, \ldots, \underline{e}_n\}$ is a spanning set.

Theorem: Every spanning set is at least as large as the basis.

Proof: Let $S = \{\underline{v}_1, \underline{v}_2, \ldots, \underline{v}_k\}$ be a spanning set and $T = \{\underline{w}_1, \underline{w}_2, \ldots, \underline{w}_{k'}\}$ be a LI set with $k < k'$.

$$\underbrace{[\underline{w}_1 \ \ \underline{w}_2 \ \cdots \ \underline{w}_{k'}]}_{W} = \underbrace{[\underline{v}_1 \ \ \underline{v}_2 \ \cdots \ \underline{v}_k]}_{V} \underbrace{\begin{bmatrix} \alpha_{11} & \alpha_{21} & \cdots & \alpha_{k'1} \\ \alpha_{21} & \alpha_{22} & & \vdots \\ \vdots & \vdots & & \vdots \\ \alpha_{k1} & \alpha_{k2} & & \alpha_{kk'} \end{bmatrix}}_{\substack{k \times k' \\ M}}$$

$$W = VM$$

$\exists \ \underline{r} \neq 0$ such that $M\underline{r} = 0$

Hence, $W\underline{r} = VM\underline{r} = 0$

Theorem: A minimum spanning set is LI.

Proof: Suppose $\dim A = k$. Let $\underline{v}_1, \underline{v}_2, \ldots, \underline{v}_k$ be

Proof.

a minimum spanning set.

If $\underline{v}_1, \underline{v}_2, \ldots, \underline{v}_k$ are linearly dependent, then

$$\underline{v}_1 = \alpha_2 \underline{v}_2 + \alpha_3 \underline{v}_3 + \cdots + \alpha_k \underline{v}_k$$

Take any $\underline{w} \in A$.

$$\underline{w} = x_1 \underline{v}_1 + x_2 \underline{v}_2 + \cdots + x_k \underline{v}_k$$

$$= x_1 (\alpha_2 \underline{v}_2 + \alpha_3 \underline{v}_3 + \cdots + \alpha_k \underline{v}_k) + x_2 \underline{v}_2 + \cdots$$

$$= (\quad) \underline{v}_2 + (\quad) \underline{v}_3 + (\quad) \underline{v}_4 + \cdots$$

$\Rightarrow \{\underline{v}_2, \underline{v}_3, \ldots\}$ is a spanning set

Basis of a subspace is

1) $\underbrace{\text{Largest}}_{\text{maximum/maximal}}$ LI set
2) Smallest spanning set

"Duality"

3) LI spanning set.

$\underline{e}_1, \underline{e}_2, \underline{e}_3$ is a basis of $\mathbb{R}^3$.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \underline{e}_1 + x_2 \underline{e}_2 + x_3 \underline{e}_3$$

Consider the subspace of $\mathbb{R}^3$ formed by

$$\left\{ \begin{bmatrix} x+y \\ y+z \\ z+x \end{bmatrix} ; \; x, y, z \in \mathbb{R} \right\}.$$

rank 3

nullity 0

$$= \underbrace{\text{im}\left(\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}\right)}$$

Rank of a matrix:

rank of a matrix is defined as $\dim(\text{im}(A))$

nullity of a matrix is defined as $\dim(\underbrace{n(A)}_{\{0\}})$

0 dimension.

$$A : \mathbb{R}^n \to \mathbb{R}^m$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix}_{3\times 2}$$

$\text{rank}(A) = 1$

basis of $\underbrace{\text{im}(A)}_{\to \mathbb{R}^3}$: $\left\{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right\}$

$\text{nullity}(A) = 1$

basis of $n(A)$:

$$\underbrace{\begin{bmatrix} -2 \\ 1 \end{bmatrix}}$$

$x + 2y = 0$

$$\boxed{\begin{matrix} x + 2y = 0 \\ 0 = 0 \end{matrix}}$$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$\text{rank}(A) = 2$

basis of $\text{im}(A) = \left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$

$x + y = 0$

$y = 0$

$z = ??$.

nullity $(A) = 1$

basis of $n(A) = \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

Properties: If S is a subspace of $\mathbb{R}^n$, then

$$\dim S \le n. \qquad \mathbb{R}^m \leftarrow \boxed{} \leftarrow \mathbb{R}^n$$

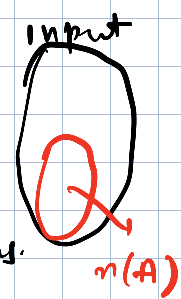
Let A be $m \times n$ matrix. $0 \le \text{rank}(A) \le m$

$$0 \le \text{nullity}(A) \le n$$

Theorem: Rank-nullity theorem. $\qquad \mathbb{R}^m \leftarrow \boxed{} \leftarrow \mathbb{R}^n$

For any $m \times n$ real matrix,

$$\underbrace{\text{rank}(A)}_{\dim(\text{Im}(A))} + \underbrace{\text{nullity}(A)}_{\dim(n(A))} = \underbrace{n}_{\text{no. of columns}}.$$



Proofs Suppose $k = \text{nullity}(A)$.

Let $\underset{\sim}{v_1}, \underset{\sim}{v_2}, \dots, \underset{\sim}{v_k}$ be a basis of $n(A)$.

$\rightarrow \mathbb{R}^n$

Expand this to a basis

$$\underline{v_1} \quad \underline{v_2} \quad \dots, \underline{v_k}, \underbrace{v_{k+1}, \underline{v_{k+2}}, \dots, \underline{v_n}}_{\rightarrow n-k}$$

of $\mathbb{R}^n$.

Any $\underline{x} \in \mathbb{R}^n$ can be written as

$$\underline{x} = \alpha_1 \underline{v_1} + \alpha_2 \underline{v_2} + \dots + \alpha_n \underline{v_n}$$

$$A \underline{x} = \alpha_1 A \underline{v_1} + \alpha_2 A \underline{v_2} + \dots + \alpha_k A \underline{v_k}$$

$$+ \alpha_{k+1} \underbrace{A \underline{v}_{k+1}} + \cdots + \alpha_n A \underline{v}_n$$

Consider the vectors $V = \{A \underline{v}_{k+1}, A \underline{v}_{k+2}, \ldots, A \underline{v}_n\}$.

V is a spanning set for $\text{im}(A)$.

Consider $\beta_{k+1} A \underline{v}_{k+1} + \beta_{k+2} A \underline{v}_{k+2} + \cdots + \beta_n A \underline{v}_n = 0$

then, $A \underbrace{\left(\beta_{k+1} v_{k+1} + \beta_{k+2} v_{k+2} + \cdots + \beta_n \underline{v}_n \right)} = 0$

$\beta_{k+1} \underline{v}_{k+1} + \beta_{k+2} \underline{v}_{k+2} + \cdots + \beta_n \underline{v}_n \in n(A)$.

$$\beta_{k+1} \underline{v}_{k+1} + \beta_{k+2} \underline{v}_{k+2} + \cdots + \beta_n \underline{v}_n$$

$$= \underbrace{\beta_1 \underline{v}_1 + \beta_2 \underline{v}_2 + \cdots + \beta_k \underline{v}_k}$$

$$\Rightarrow \beta_1 = \beta_2 = \cdots = \beta_{k+1} = \cdots = \beta_n = 0.$$

Hence vectors in V are LI. Since V is LI spanning set for $\text{im}(A)$, V must be a basis.

Hence $\text{rank}(A) = \dim(\text{im}(A)) = n - k$
$= n - \text{nullity}(A)$.

"Dimension-conserving" principle

$$\underbrace{\text{rank}(A)} \leftarrow \boxed{} \leftarrow \underbrace{n - \text{nullity}(A)}_{\text{input dimensions.}}$$

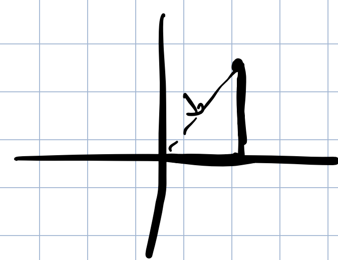
Orthogonality:

(Inner product) $\underline{v_1}, \underline{v_2} \in \mathbb{R}^n$

$\underbrace{\underline{v_1}^T v_2}$ is defined as inner product of $\underline{v_1}$ & $\underline{v_2}$.

$$\underline{v_1^T v_2} = v_{11}v_{21} + v_{12}v_{22} + v_{13}v_{23} + \cdots$$

$$\underline{v_1} = \begin{bmatrix} v_{11} \\ v_{12} \\ \vdots \end{bmatrix} \qquad \underline{v_2} = \begin{bmatrix} v_{21} \\ v_{22} \\ v_{23} \\ \vdots \end{bmatrix}$$



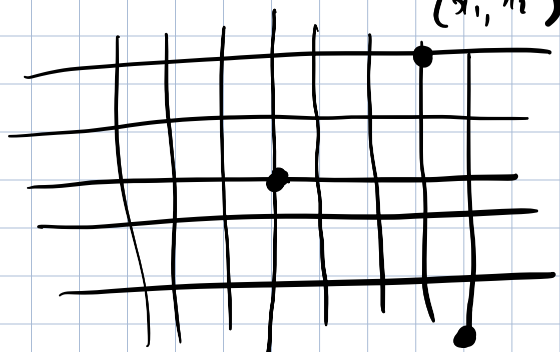
Norm:

$\underline{v} \in \mathbb{R}^n$

$$\|\underline{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

(Pythagoreas thm)

Euclidean distance.



$$\|\underline{v}\| = |v_1| + |v_2| + |v_3| + \cdots + |v_n|$$

"Distance"

$|x_1| + |x_2|$

Manhattan distance.

$x_1 \quad x_2$
$y_1 \quad y_2$

$|x_1 - y_1|$

Definition: Norm

$$\|\ \| : \mathbb{R}^n \to \mathbb{R}$$

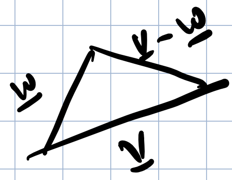
$\|\ \|$ is a norm if

① $\|\underline{v}\| \geq 0$ for all $\underline{v} \in \mathbb{R}^n$

② $\|\alpha \underline{v}\| = |\alpha| \|\underline{v}\|$ for all $\underline{v} \in \mathbb{R}^n$, $\alpha \in \mathbb{R}$.

③ $\|\underline{v}\| = 0$ if & only if $\underline{v} = 0$.

④ Triangle inequality



$$\|\underline{v} + \underline{w}\| \leq \|\underline{v}\| + \|\underline{w}\| .$$

Example: I $\|\underline{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$ is called the Euclidean-norm 2-norm.

satisfies ①, ②, ③.

To check ④,

$$\|\underline{v} + \underline{w}\|^2 = (v_1 + w_1)^2 + (v_2 + w_2)^2 + \cdots + (v_n + w_n)^2$$

$$= v_1^2 + v_2^2 + \cdots + v_n^2 + w_1^2 + w_2^2 + \cdots + w_n^2$$
$$+ 2v_1w_1 + 2v_2w_2 + \cdots$$

To show:

$$\leq \|\underline{v}\|^2 + \|\underline{w}\|^2 + 2\|\underline{v}\|\|\underline{w}\|$$

$$v_1^2 + v_2^2 + \cdots + v_n^2 + w_1^2 + w_2^2 + \cdots + w_n^2$$
$$+ 2\sqrt{w_1^2 + \cdots w_n^2}\sqrt{v_1^2 + v_2^2 + \cdots v_n^2}$$

$$v_1w_1 + v_2w_2 + \cdots + v_nw_n$$

$$\leq \sqrt{v_1^2 + v_2^2 + \cdots}\sqrt{w_1^2 + w_2^2 + \cdots}$$

Cauchy-Schwarz inequality.

Example II:

$$\|\underline{v}\| = |v_1| + |v_2| + |v_3| + \cdots +$$

1-norm.

① $\|\underline{v}\| \geq 0$ True Non-negativity

② $\|\alpha \underline{v}\| = |\alpha| \|\underline{v}\|$ True Homogeneity

if $\underline{v} = 0$. Positive -

③ $\|\underline{v}\| = 0$ if & only if $\underline{v} = 0$ definiteness.

④ $\underbrace{\|\underline{v} + \underline{w}\|}_{}$ $\leq$ $\|\underline{v}\| + \|\underline{w}\|$ Triangle inequality

$\underbrace{|v_1 + w_1|} + |v_2 + w_2| + \cdots \leq |v_1| + |w_1| + |v_2| + |w_2| + \cdots$

$$\underbrace{|v_1 + w_1| \leq |v_1| + |w_1|}$$

Excercise: $\|\underline{v}\|_\infty = \max_{i=1}^{n} |v_i| \longrightarrow$ Is this a norm?

$\longrightarrow \|\underline{v}\|$ = number of non-zero entries in $\underline{v}$.

Is this a norm?

Non-negativity ✓

Homogenity ✗

definiteness ✓

Triangle inequality ✓

$|x+y| \leq |x| + |y|$

$\|v + w\| \leq \|v\| + \|w\|$

nonzero entries in $v+w$ | non-zero entries in v | nonzero entries in w.

$\|\underline{v}\|_\infty = \max |v_i|$

non-negative ✓

homogeneous ✓

definite ✓

triangle inequality ✓

$\underline{v} \nearrow \begin{matrix} v_1 \\ v_2 \\ \vdots \\ | \end{matrix}$ $\quad \underline{w} \nearrow \begin{matrix} w_1 \\ w_2 \\ \vdots \\ | \end{matrix}$

$\|\underline{v} + \underline{w}\|_\infty = \max \underbrace{|v_i + w_i|}$

$|v_1+w_1| \qquad |v_1|+|w_1| \le |v_i|+|w_i|$

$|v_2+w_2| \qquad |v_2|+|w_2|$

$\vdots \qquad \vdots$

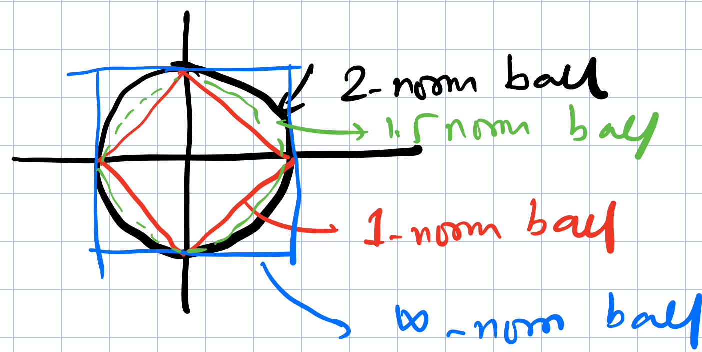
$|v_n+w_n| \qquad |w_n|+|v_n|$

Norm-ball

2-norm ball of radius 1

$$\{\underline{x} : \|\underline{x}\|_2 = 1\}$$

$\mathbb{R}^2$ $\quad \{\underline{x} : x_1^2 + x_2^2 \le 1\}$



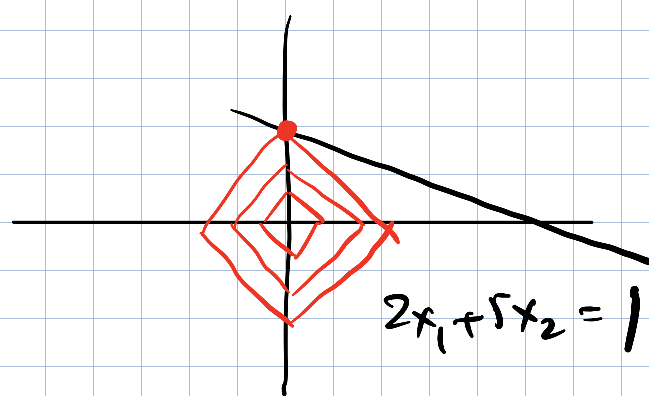
1-norm ball of radius 1

$$\{\underline{x} : |x_1| + |x_2| \le 1\}$$

∞-norm ball of radius 1

$$\{\underline{x} : \max\{|x_1|, |x_2|\} \le 1$$

$x_1 + 2x_2 = 1$



$\boxed{\text{min-norm problem}}$

$$\min \|\underline{x}\|_2$$
$$\text{s.t.} \quad 2x_1 + 5x_2 = 1$$

Solving $\min \|\underline{x}\|_1$ promotes "sparsity".

$\text{s.t.} \quad 2x_1 + 5x_2 = 1$

Solving $\min \|\underline{x}\|_\infty$

$\text{s.t. } 2x_1 + 5x_2 = 1$ promotes a solution with equal values on x_1 & x_2.

ℓ ??

ℓ_p norm: (p-norm)

$$\|\underline{x}\|_p = \left(|x_1|^p + |x_2|^p + \cdots + |x_n|^p\right)^{1/p}$$

Satisfies all 4 properties of norm for $p \geq 1$.

$p = 1$

$p = 2$

$p = \infty$.

Norms $\leftrightarrow$ Distance.

$\|x - y\|_q$

The 2-norm is connected to the concept of inner-product

Inner products: $\longleftrightarrow$ "Angle".

$\mathbb{R}^2$

$(x_1 \quad x_2)$

$(y_1 \quad y_2)$

(x, y)

$\langle \underset{\mathbb{R}^n}{x}, \underset{\mathbb{R}^n}{y} \rangle$

$\mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}$

1. $\langle x, \alpha y \rangle = \alpha \langle x, y \rangle = \langle \alpha x, y \rangle$ Homogenity
2. $\langle x, y_1 + y_2 \rangle = \langle x, y_1 \rangle + \langle x, y_2 \rangle$ Additivity
3. $\langle x, y \rangle = \langle y, x \rangle$ symmetry
4. $\langle x, x \rangle \geq 0$ positivity
and $\langle x, x \rangle = 0$ only when $x = 0$.

$\sqrt{\underbrace{\langle x, x \rangle}}$

Standard inner product on $\mathbb{R}^n$ is given by

$$\langle \underline{x}, \underline{y} \rangle = \underline{x}^T \underline{y} = x_1 y_1 + x_2 y_2 + \dots + x_n y_n.$$

Exercise : Verify this satisfies all 4-properties of inner product.

$$\langle \underline{x}, \underline{y} \rangle = w_1 x_1 y_1 + w_2 x_2 y_2 + w_3 x_3 y_3 + \dots$$

$$(w_i \geq 0)$$

Every inner product "induces" a norm on $\mathbb{R}^n$.

For standard inner product :

$$\sqrt{\underline{x}^T \underline{x}} = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} = \|\underline{x}\|_2$$

$$\underbrace{[x_1 \ x_2 \ \dots \ x_n]}_{\underline{x}^T} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{x}$$

Parallelogram law :

$$\|\underline{x} - \underline{y}\|_2^2 + \|\underline{x} + \underline{y}\|_2^2 = 2\left(\|\underline{x}\|_2^2 + \|\underline{y}\|_2^2\right)$$

Proof : $(\underline{x} - \underline{y})^T (\underline{x} - \underline{y}) + (x+y)^T (x+y)$

$$= (\underline{x}^T - \underline{y}^T)(\underline{x} - \underline{y}) + (\underline{x}^T + \underline{y}^T)(\underline{x} + \underline{y})$$

$$= \underline{x}^T \underline{x} + \underline{y}^T \underline{y} \cancel{- 2x^T y} \qquad \underline{x}^T \underline{x} + \underline{y}^T \underline{y} \cancel{+ 2\underline{x}^T \underline{y}}$$

$$= 2\left(\|\underline{x}\|^2 + \|\underline{y}\|^2\right)$$

Exercise: Do 1-norm or ∞-norm satisfy the parallelogram law?.
(Check a few examples)

Cauchy-Schwarz inequality

$$-\|\underline{x}\|\|\underline{y}\| \le \langle x, y \rangle \le \|\underline{x}\| \|\underline{y}\|$$

where $\| \ \|$ is the norm induced by $\langle \ \rangle$

$$-\|\underline{x}\|_2\|\underline{y}\|_2 \le \underline{x}^T\underline{y} \le \|\underline{x}\|_2 \|\underline{y}\|_2$$

Proof: Suppose $\|\underline{x}\|_2 = \|\underline{y}\|_2 = 1$.

$$0 \le \|\underline{x}-\underline{y}\|_2^2 = (\underline{x}-\underline{y})^T(\underline{x}-\underline{y})$$
$$= \|\underline{x}\|_2^2 + \|\underline{y}\|_2^2 - 2\underline{x}^T\underline{y}$$
$$= 2(1-\underline{x}^T\underline{y})$$
$$\Rightarrow \underline{x}^T\underline{y} \le 1 \quad \text{and} \quad \underline{x}^T\underline{y} = 1$$
only when $\underline{x} = \underline{y}$

$$0 \le \|\underline{x}+\underline{y}\|_2^2 = 2\underbrace{(1+\underline{x}^T\underline{y})}$$
$$-1 \le \underline{x}^T\underline{y} \quad \text{and} \quad \underline{x}^T\underline{y} = 1$$
only when $\underline{x} = -\underline{y}$.

Suppose $\underline{x}$ and $\underline{y}$ are general vectors. $(\underline{x}, \underline{y} \neq 0)$

$$\tilde{\underline{x}} = \frac{\underline{x}}{\|\underline{x}\|_2}$$ ← Normalization. $\|\tilde{x}\|_2 = 1$.

$$\tilde{\underline{y}} = \frac{\underline{y}}{\|\underline{y}\|_2}$$

$$-1 \leq \tilde{\underline{x}}^T \underline{y} \leq 1$$

$$-1 \leq \left(\frac{\underline{x}}{\|\underline{x}\|_2}\right)^T \frac{\underline{y}}{\|\underline{y}\|_2} \leq 1$$

$$\Rightarrow \quad -\|\underline{x}\|_2 \|\underline{y}\|_2 \leq \underline{x}^T y \leq \|\underline{x}\|_2 \|\underline{y}\|_2 .$$

Average ? n-numbers

$$\frac{x_1 + x_2 + \cdots + x_n}{n}$$

$$= \frac{1}{n} \mathbf{1}^T \underline{x}$$

all 1 vector

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \end{bmatrix}$$

Taylor's Series:

$$f(x) \approx f(0) + x f'(0)$$

$$f(x_1, x_2) \approx f(0,0) + \frac{\partial f}{\partial x_1}(0,0)\, x_1$$

∂x_1

$$+ \frac{\partial f}{\partial x_2}(0,0) \quad x_2$$

(Gradient descent)

$$= f(0,0) + \underline{\nabla f}^T \underline{x}$$

$$\underline{\nabla f} = \begin{bmatrix} \frac{\partial f}{\partial x_1}(0,0) \\ \frac{\partial f}{\partial x_2}(0,0) \end{bmatrix}$$

Gradient.

$f(x_1, x_2)$: cost of moving to x_1, x_2.

from origin, which x_1, x_2 should I move to so that I incur the least cost (assume x_1, x_2 are small)

<u>Soln</u> By Cauchy Schwarz, move in negative gradient direction.

<u>Defn</u>: we say that $\underline{x}$ $\underline{y}$ are orthogonal if $\underline{x}^T \underline{y} = 0$.

<u>Proof</u>: $\|\underline{x} + \underline{y}\|_2^2 = \|\underline{x}\|_2^2 + \|\underline{y}\|_2^2$ if and only if $\underline{x}^T \underline{y} = 0$.

$\underline{x}, \underline{y} \in \mathbb{R}^n \qquad \langle \underline{x}, \underline{y} \rangle = \underline{x}^T \underline{y}.$

$$\boxed{\underline{x}^T \underline{x} = \|\underline{x}\|_2^2}$$

Property: If $\underline{x}$ and $\underline{y}$ are non-zero orthogonal, then $\{\underline{x}, \underline{y}\}$ are linearly independent.

Proof:

$$\alpha_1 \underline{x} + \alpha_2 \underline{y} = 0$$

$$\underline{x}^T(\alpha_1 \underline{x} + \alpha_2 \underline{y}) = 0$$

$$\underbrace{\alpha_1 \underline{x}^T \underline{x}} + \alpha_2 \cancel{\underline{x}\underline{y}^T} = 0 \Rightarrow \alpha_1 = 0$$

Similarly, multiply by $y^T \rightarrow \alpha_2 = 0$

$\underline{x_i} \in \mathbb{R}^n$

Let $\underline{x_1}$ $\underline{x_2}$, ..., $\underline{x_k}$ be non-zero orthogonal vectors.

then $\{\underline{x_1}, \underline{x_2}, \ldots, \underline{x_k}\}$ is linearly independent.

(orthogonal set of vectors)

$$V = [\underline{x_1} \ \underline{x_2} \ \cdots \ \underline{x_k} \qquad k \leq n$$

Suppose in addition that $\|\underline{x}_1\|_2 = \|x_2\|_2 = \cdots = \|x_k\|_2 = 1$

(orthonormal set of vectors)

orthogonality of $\underline{x_1}, \underline{x_2}$ -- can also be stated as (tall)

$$V^T V = I$$

Gram matrix

For any matrix A, $(A^TA)_{ij}$ is the inner product of i^{th} column of A with j^{th} column of A.

$(AB)_{ij}$ is inner product of i^{th} rows of A with j^{th} column of B.

Orthogonal matrix:

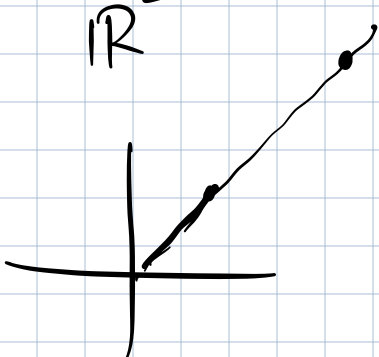
We say that $V_{n \times K}$ $(K \leq n)$ is an orthogonal matrix if $V^TV = I$.

$$\leftarrow \boxed{V} \leftarrow \underline{x}$$

How is $\|V\underline{x}\|_2$ related to $\|\underline{x}\|$?

$$\|V\underline{x}\|_2^2 = (V\underline{x})^T V\underline{x}$$

$\mathbb{R}^2$



$$= \underline{x}^T V^T V \underline{x} = \underline{x}^T\underline{x}$$

$$= \|\underline{x}\|_2^2$$

$$\|V\underline{x}\|_2 = \|\underline{x}\|_2 .$$

orthogonal matrices

$\updownarrow$

reflections & rotations

Summary Suppose $x_1, x_2, \ldots, x_K \in \mathbb{R}^n$ are orthonormal.

then

① $\underline{x}_1, \underline{x}_2, \ldots, \underline{x}_k$ are LI

② $V = [\underline{x}_1, \underline{x}_2, \ldots, \underline{x}_k]$ is an orthogonal matrix, it satisfies $V^TV = I$ and $\|V\underline{x}\|_2 = \|\underline{x}\|_2$ for any $\underline{x}$.

③ If the system $\underline{y} = V\underline{x}$ has a solution, it is given by $\underline{x} = V^T\underline{y}$.

"Least squares"

$$\underline{y} = x_1\underline{v}_1 + x_2\underline{v}_2 + \cdots + x_k\underline{v}_k$$

where $\underline{v}_1, \underline{v}_2, \underline{v}_3, \ldots, \underline{v}_k$ are orthonormal

then $x_1 = \underline{v}_1^T\underline{y}$, $x_2 = \underline{v}_2^T\underline{y}$, $x_3 = \underline{v}_2^T\underline{y}$, ..., $x_k = \underline{v}_k^T\underline{y}$

Consider the case

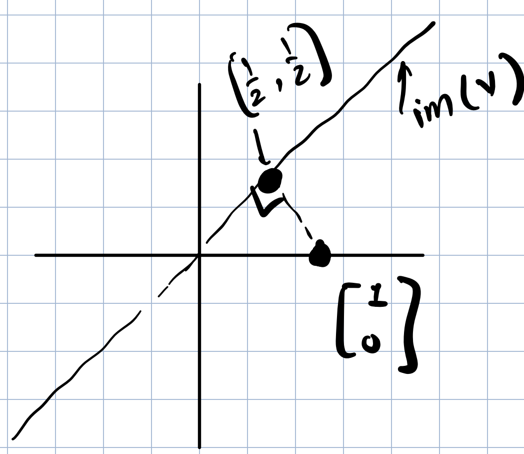
$\underline{K=1}$

Here V is a unit-norm (column) vector.

$$\underset{n\times 1}{y} = \underset{n\times 1}{V}\,\underset{1\times 1}{x}$$

$n=2$

$$V = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \qquad y = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$



$$VV^T = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$y = V\underline{x}$$

$$V^Ty = \left(\frac{1}{\sqrt{2}} \;\; \frac{1}{\sqrt{2}}\right)\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$x = V y$

$$x^* = \frac{1}{\sqrt{2}}$$

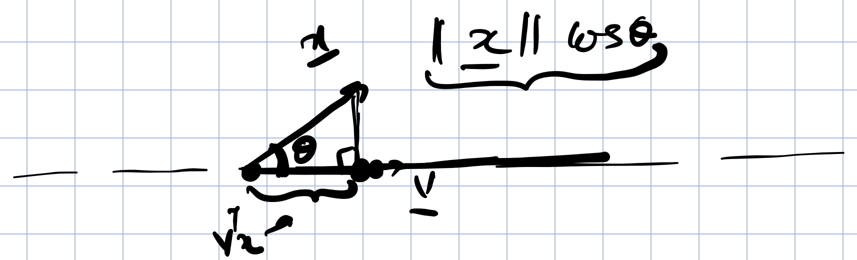
$\underline{y} \longleftrightarrow VV^T \underline{y} \qquad V x^*$

$$VV^T \underline{y} = V x^* = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \frac{1}{\sqrt{2}} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

Projection: projection of $\underline{x} \in \mathbb{R}^n$ along span$\langle \underline{v} \rangle$ is defined as the point in span$\langle \underline{v} \rangle$ that is closest to $\underline{x}$.

$\|\underline{v}\|_2 = 1$

$$\text{proj}(\underline{x}) = \alpha^* \underline{v} \quad \text{where} \quad \alpha^* = \underbrace{\underset{\alpha \in \mathbb{R}}{\text{argmin}} \|\underline{x} - \alpha \underline{v}\|_2}$$



$$\cos\theta = \frac{\text{base}}{\|x\|}$$

Find α for which $\underbrace{\|\underline{x} - \alpha \underline{v}\|_2^2}$ is smallest.

$$(\underline{x} - \alpha \underline{v})^T (\underline{x} - \alpha \underline{v})$$

$$= \underbrace{\underline{x}^T \underline{x} + \alpha^2 \underline{v}^T \underline{v} - 2 \alpha \underline{v}^T \underline{x}}$$

$$\alpha^* = \underbrace{\underline{v}^T \underline{x}}$$

$\text{proj}(\underline{x}) = (\underline{v}^T \underline{x})\, \underline{v}$ (projection of $\underline{x}$ along $\underline{v}$)

$$\underline{v}^T \underline{x} \longleftrightarrow \|\underline{x}\|_2 \cos\theta$$

linear algebra $\qquad\qquad\qquad\qquad$ geometric

$\|v\|_2 = 1 \qquad \dfrac{\underline{v}^T \underline{x}}{\|\underline{x}\|} \longleftrightarrow \cos\theta$

K = 2

Projection of $\underline{x}$ along the plane spanned by $\underline{v}_1, \underline{v}_2$ ($\underline{v}_1, \underline{v}_2$ are orthonormal)

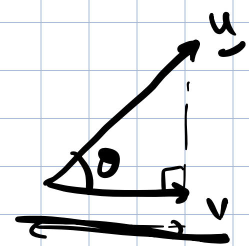
is given by $\underbrace{(\underline{v}_1^T \underline{x})\, \underline{v}_1 + (\underline{v}_2^T \underline{x})\, \underline{v}_2}_{= V V^T \underline{x}}$

Property (arbitrary K).

Projection of a vector $\underline{x}$ along the $\mathrm{im}(V)$ (V is orthogonal matrix)

is given by $V V^T \underline{x}$.

Let $\underline{u}$ be a vector and $\underline{v}$ a unit vector. The projection of $\underline{u}$ along $\underline{v}$ is $\underbrace{(\underline{u}^T \underline{v}) \; \underline{v}}$



Revisiting rank

$$\text{rank}(A) = \dim(\text{im}(A))$$

$=$ largest number of linearly independent columns in A.

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

$$\text{nullity}(A) = \dim(\text{null}(A))$$

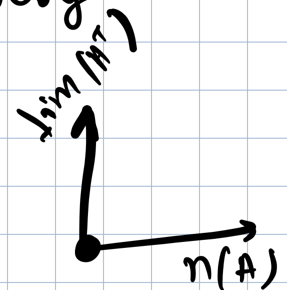
$$\text{rank}(A) + \text{nullity}(A) = \text{no of column of } A.$$

$\underline{x} \in n(A)$: $\underbrace{A\,\underline{x}} = 0$

$\underline{x}$ is perpendicular to all rows of A.

Consider the $\text{im}(A^T)$: space spanned by rows of A.

Every vector in $n(A)$ is orthogonal to every vector in $\mathrm{im}(A^T)$.



$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\mathrm{im}(A^T) = \left\{ \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\}$$

$$= \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \\ 2x_1 \end{bmatrix}$$

Property: $A\underline{x} = 0 \iff A^TA\underline{x} = 0.$

(related to orthogonality of null space & image space)

if & only if

$$\boxed{A A^T \underline{x} = 0 \iff A^T \underline{x} = 0}$$

✓

$$\underline{A\underline{x} = 0} \Rightarrow \underbrace{A^TA\underline{x} = 0}$$

$$\underbrace{A^TA\underline{x} = 0} \Rightarrow A\underline{x} = 0$$

$$\underbrace{A^TAx = 0}, \quad Ax = b$$

$$\underline{x}^T A^T A \underline{x} = \underbrace{\underline{x}^T 0}$$

$$\Rightarrow \|Ax\|_2^2 = 0 \Rightarrow A\underline{x} = 0$$

$A: m \times n$

Property: $\text{rank}(A) = \text{rank}(A^T)$.

$\underbrace{n(A)} \subseteq \mathbb{R}^n$

$\text{im}(A^T) \subseteq \mathbb{R}^n$

Let $\underline{y}_1, \underline{y}_2, \ldots, \underline{y}_r$ be a basis of $\text{im}(A^T)$.

$\in \mathbb{R}^n$ $\qquad r = \text{rank}(A^T)$

$\underbrace{A\underline{y}_1, \quad A\underline{y}_2, \quad \ldots, A y_r}$ are in $\text{im}(A)$.

Are $A\underline{y}_1, A\underline{y}_2, \ldots, A\underline{y}_r$ LI?

Suppose they are LD. Then,

$\alpha_1 A\underline{y}_1 + \alpha_2 A\underline{y}_2 + \cdots + \alpha_r A y_r = 0$

$A(\underbrace{\alpha_1 \underline{y}_1 + \alpha_2 \underline{y}_2 + \cdots + \alpha_r \underline{y}_r}) = 0$

$A(A^T(\alpha_1 z_1 + \alpha_2 z_2 + \cdots + \alpha_r z_r)) = 0$

$$\boxed{A \begin{bmatrix} y_1 & y_2 & \cdots & y_r \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_r \end{bmatrix}}$$

$A = \begin{bmatrix} 1 & 1 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\quad \text{im}(A^T) = \begin{bmatrix} k \\ k \end{bmatrix}$

$\underline{y}_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$

$\underline{y}_2 = \begin{bmatrix} 0, & 1 \end{bmatrix}^T$

LI

$A y_1 = 1 \qquad A y_2 = 1$

$y_1 = A^T z_1, \quad y_2 = A^T z_2, \ldots$

$\underbrace{\alpha_1 \underline{y}_1 + \alpha_2 \underline{y}_2 + \cdots + \alpha_r \underline{y}_r}$
$\in n(A)$.

But, by construction, $\alpha_1 \underline{y}_1 + \alpha_2 \underline{y}_2 + \cdots + \alpha_r \underline{y}_r \in \text{im}(A^T)$

therefore, $\alpha_1 \underline{y}_1 + \alpha_2 \underline{y}_2 + \cdots + \alpha_r \underline{y}_r = 0$.

hence $\alpha_1 = \alpha_2 = \cdots = \alpha_r = 0$.

We conclude that.

$$\text{rank}(A) \geq \text{rank}(A^T).$$

Similarly, $\text{rank}(A^T) \geq \text{rank}(A)$

Hence $\text{rank}(A) = \text{rank}(A^T)$.

$\text{rank}(A^T)$ is also the number of LI rows in A -

Going back,

$$\underbrace{\text{rank}(A)}_{\text{no of dimensions along rows of A}} + \underbrace{\text{nullity}(A)}_{\text{no of dimensions perpendicular to rows of A}} = n.$$

Consider $A = \underline{a}^T$ (Single row)

$\underline{a} \neq 0$.

$\text{rank}(A) = 1$

$\text{nullity}(A) = n-1$.

$n(A) = \{\underline{x} : \underline{a}^T \underline{x} = 0\}$

$A\underline{x} = \underline{b}$ $\quad$ $\underbrace{A : m \times n}$

$\text{rank}(A) \leq n$

Suppose

$\text{rank}(A) = m$ (rows LI)

$\text{rank}(A) \leq m$

then what can be said about $A\underline{x} = \underline{b}$? (at least one solution)

A is onto, $A\underline{x} = \underline{b}$ has atleast one solution.

Suppose $\text{rank}(A) = n$ (columns linearly ind)

$A\underline{x} = \underline{b}$ has atmost one solution.

A is one-one.

If A is square, $\text{rank}(A) = n$ (rows are LI columns are LI).

then A is one-one onto.

$$\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{100} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$A\underline{x} = \underline{b}$

measured

$x_1 = b_1$

$x_2 = 100 b_2$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Singular / Eigen values of the matrices

$$\underline{x}(t+1) = A\,\underline{x}(t)$$

Linear Dynamical system.

$x(0)$

$Ax(0)$

$A^2 x(0)$

$A^n x(0)$

What happens to A^n as $n \to \infty$?

$\underline{A x} \leftarrow \square \leftarrow \underline{x}$

Amplification: $\max \dfrac{\|A\underline{x}\|}{\|\underline{x}\|}$

$\min \dfrac{\|A\underline{x}\|}{\|\underline{x}\|}$

Computational / numerical Linear Algebra

(QR Decomposition)

$A : m \times n$

$A\underline{x}$ → How costly is this multiplication?

$\underline{x} \in \mathbb{R}^n$

└ Number of arithmetic operations
addition, multiplication.

Inner product: $\underline{x}$ $\underline{y}$

$$\langle \underline{x}, \underline{y} \rangle = \underline{x}^T \underline{y} = \underbrace{x_1 y_1 + x_2 y_2 + \cdots + x_n y_n}$$

n multiplication

$n-1$ add.

$\underbrace{2n-1}$

$2n$

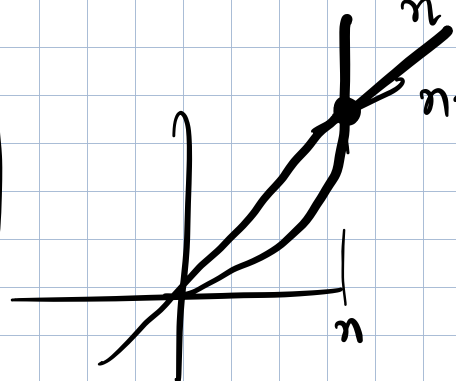
$3n-1$

$10n$

$1000n$

$1000n$

n^2



Big-Oh notation:

We say that $f(n) = O(g(n))$ if

"order"

$f(n) \le \underline{(\text{Const})\, g(n)}$ for all n large enough.

$$2n-1 = O(n)$$

$$\underbrace{2n-1}_{f(n)} \le 2(\underbrace{n}_{g(n)})$$

Number of arithmetic operations in computing the inner product is $O(n)$.

A $m \times n$ $\quad \underline{x} \in \mathbb{R}^n$.

Number of arithmetic operations for computing $A\underline{x}$ is $m(2n-1) = O(nm)$

$$\underbrace{2nm} - \underset{\geq}{m} \le 2nm$$

$$= O(nm).$$

$A: n \times n$ $\quad \underbrace{O(n^2)}$

If A is all 1 matrix, $A\underline{x}$ can be computed in $O(n)$.

Number of arithmetic operations to comput

$$\underset{m\times n}{A}\ \underset{n\times p}{B} = A\,[\underline{b}_1\ \underline{b}_2\ --\ \underline{b}_p]$$

$m\times p$

$$= [A\underline{b}_1\ \ A\underline{b}_2\ --\ A\underline{b}_p]$$

$$O(mnp)$$

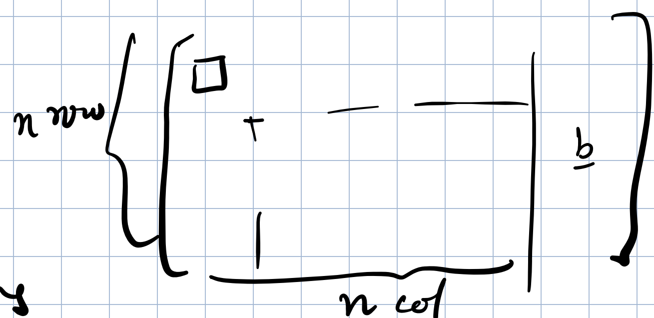
<u>Excercise</u>: A is $n\times n$ matrix

(i) $A\underline{x} = \underline{b}$ by row reduction

find number of arithmetic operations

(ii) $\det(A)$: number of arithmetic operation?
$n\times n$

To make entries in the first column zero, we need $2(n-1)n$ operations



<u>2nd col</u>: $2(n-2)(n-1)$

$$\left|2\left[\underbrace{n(n-1)} + (n-1)(n-2) + (n-2)(n-3) + \cdots\right]\right|$$

$$\le \left|\left[\underbrace{n\cdot n^2} + (c_1\boxed{n})\,n + c_2 n\right]\right|$$

$$O(n^3)$$

$$\begin{bmatrix} 1 & 5 & 15 \\ 0 & 2 & 20 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

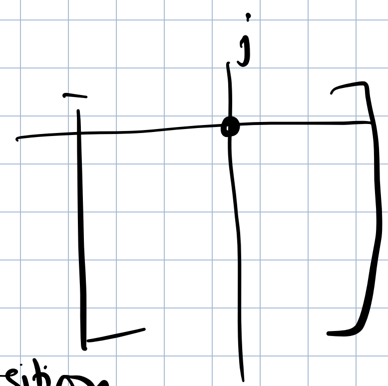
Back-substitution.

forward-substitution

$$\begin{bmatrix} 1 & 0 & 0 \\ 5 & 20 & 0 \\ 15 & 20 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_2 \end{bmatrix} \qquad O(n^2)$$

$$\det A = -\sum (-1)^j \underbrace{\det |A_{1j}|}_{} \quad \text{— (1)}$$

obtained by removing entry in $(1,j)^{th}$ position.



Let the complexity of finding $n \times n$ det be $g(n)$ using (1) above.

$g(2) =$

$$g(n) = n\, g(n-1) + n \qquad g(1) = 1$$

$$g(n) \approx n!$$

$$n^{n/2} \le n! \le n^n$$

$1 \cdot 2 \cdot 3 \cdots n$

$$n^n \le (n!)^2$$

$n!$ $1 \cdot 2 \cdot 3 \cdots n$

$n!$ $n(n-1)(n-2)\cdots 1$

$$(n!)^2 \ge n^n$$

$2(n-1) > n$

$n > 2$

$$\underbrace{R_i \leftarrow R_i + \sigma R_j}$$

Determinant can be computed via row reduction in $O(n^3)$.

Determinant of a square matrix

$\det(A) = f(\underline{a_1}, \underline{a_2}, \dots, \underline{a_n})$ where $\underline{a_1}, \underline{a_2}, \dots, \underline{a_n}$ are columns of A.

(i) $f(c\,a_1, a_2, \dots, a_n) = c\,f(a_1, a_2, \dots, a_n)$

$f(a_1 + b_1, a_2, \dots, a_n) = f(a_1, a_2, \dots, a_n) + f(b_1, a_2, \dots, a_n)$

(multilinear functions)

$f(x, y) = \underline{\underline{xy}}$

(ii) $f(a_2, a_1, a_3, \dots, a_n) = -f(a_1, a_2, \dots, a_n)$

alternating.

(iii) $f(e_1, e_2, \dots, e_n) = 1$.

Determinant is the <u>only</u> function that satisfies the above.

Tr = sum of diagonal entries.

<u>Trace</u>: $Tr(A)$ satisfies the following properties

(i) $Tr(A+B) = Tr(A) + Tr(B)$, $Tr(cA) = c\,Tr(A)$

(ii) $Tr(AB) = Tr(BA)$ for any A, B.

$A: 1 \times n$ $B: n \times 1$

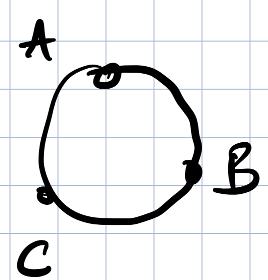
$AB: 1 \times 1$ $BA: n \times n$

(iii) $Tr(I) = n$.

Trace is the only function satisfying these properties

$$Tr(ABC) = Tr(CAB) = Tr(BCA)$$

cyclic property



$$\underbrace{Tr(\underline{x}\,\underline{x}^T)} = \|\underline{x}\|_2^2$$

$\underline{x} \in \mathbb{R}^n$ $\longrightarrow Tr(\underline{x}^T\underline{x}) = \underline{x}^T\underline{x}$

Let $\|A\|_F^2$ = sum of squares of all entries in A.

(Frobenius)

$$\|A\|_F^2 = \mathrm{Tr}(A^T A) = \mathrm{Tr}(A A^T)$$

Matrix decompositions / factorization

LU decomposition

row operation

column op

$$P_1 A Q_2 = B$$

original final

$$\underbrace{A}_{\text{original}} = P_1 B P_2$$

final.

① LU decomposition: $A = \underbrace{L}_{\text{lower } \Delta r} \; \underbrace{U}_{\text{upper } \Delta r}$

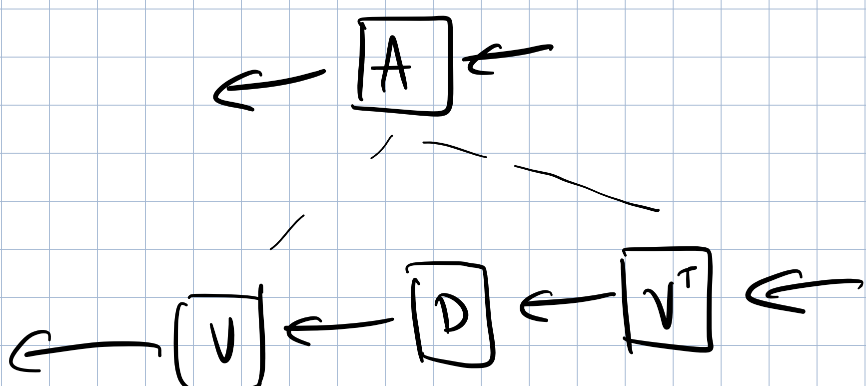
$O(n^3)$

② Spectral decomposition:

Any symmetric matrix can be written as

$O(n^3)$ $\qquad \underbrace{A = VDV^T}$ where V is orthogonal, D is diagonal.

rotations reflections



(3) QR decomposition

$O(n^3)$.